HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Analyzed linguistic datasets using **Pandas** and **regular expressions** to identify patterns and errors in Text-to-Speech outputs for a global social networking client.
- Developed and maintained text normalization rules and test sets in Python, ensuring accuracy against defined linguistic standards for new voice deployments.
- Evaluated the linguistic and audio quality of Text-to-Speech outputs, identifying and resolving language-specific challenges such as word stress and inflection for Italian and Spanish.
- Contributed to improving the TTS pipeline by supporting text normalization and pronunciation prediction processes, analysing data quality and proposing test scenarios.
- Implemented Python scripts to automate data quality checks and linguistic investigations, processing approximately **5,000 text samples** daily for iterative quality improvement.
- Collaborated with operations teams to investigate and resolve language-related issues reported through the Linguist Runbook, ensuring consistent voice quality post-launch.
- Containerized data analysis scripts with **Docker** and managed deployments via **Jenkins** pipelines, ensuring reproducible and efficient execution of linguistic quality checks.

SKILLS

Languages & Tools – Python, Pandas, Regular Expressions (Regex), SQL, Java, Spring Boot, Docker, Linux, Git, Jenkins

Speech Technologies – Text-to-Speech (TTS), Text Normalization, Pronunciation Prediction, Grapheme-to-Phoneme (G2P) evaluation

Data Analysis & Linguistics – Linguistic Analysis, Data Quality Checks, Dataset Analysis, Data Annotation, IPA (International Phonetic Alphabet)

Cloud & CI/CD – AWS (EC2, S3, Lambda, CloudWatch), Jenkins

Databases – PostgreSQL, MySQL, Oracle

PROJECTS

TTS Linguistic Quality Assurance Framework

- Developed a Python framework to automate the evaluation of Text-to-Speech outputs, processing over **10,000 text samples** to identify linguistic discrepancies.
- Implemented text normalization rules and validation scripts using **Pandas** and **regex** to ensure phonetic accuracy and adherence to language-specific grammar.
- Created regression test sets for Italian and Spanish locales, improving the detection rate of pronunciation errors by approximately **30%**.

Automated Data Processing Pipeline for Speech Datasets

- Engineered a Python-based ETL pipeline to clean and structure raw speech datasets for training TTS models, handling datasets up to **1.5 GB**.
- Applied advanced text processing techniques, including segmentation and inflection analysis, to prepare datasets for pronunciation prediction modules.
- Integrated the pipeline with **AWS S3** for scalable storage and retrieval, reducing manual data preparation time by **40%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)